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Joint mechanism and multi-joint deviced

Abstract

Without an increase in the cost, a movable range is defined. A first joint mechanism includes a base part, and a rotary part that rotates relatively to the base part. The first joint mechanism includes a regulated part that rotates together with the rotary part, a regulating part that is disposed on the extension of the rotation locus of the regulated part, and that has a function of regulating rotation of the regulated part relative to the base part, within a first movable range, and a movable range defining member that is disposed on either the base part or the rotary part, and that defines, as a movable range of the regulated part, a second movable range that is narrower than the first movable range.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a joint mechanism and a multi-joint device.

BACKGROUND ART

(2) For example, PTL 1 discloses a robot that has joints.

CITATION LIST

Patent Literature

(3) [PTL 1] Japanese Patent Laid-Open No. 2003-117858

SUMMARY

Technical Problem

(4) In order to prepare joint mechanisms having different movable ranges, it is necessary to separately manufacture the mechanisms according to the respective movable ranges. In addition, a part for regulating the movable range may be worn out and deteriorate due to long time use of the joint mechanism. If so, it is necessary to change the joint mechanism completely. Moreover, in a case where a plurality of joint mechanisms are used in one device, it is necessary to prepare a plurality of joint mechanisms having different movable ranges suitable for respective use positions or for respective purposes of use. These circumstances should bring about a cost increase.

Solution to Problem

(5) The present disclosure proposes a joint mechanism including a base part and a rotary part that rotates relatively to the base part. The joint mechanism includes a regulated part that rotates together with the rotary part, a regulating part that is disposed on an extension of a rotation locus of the regulated part, and has a function of regulating rotation of the regulated part relative to the base part, within a first movable range, and a movable range defining member that is mounted on either the base part or the rotary part, and defines, as a movable range of the regulated part, a second movable range that is narrower than the first movable range. This joint mechanism can define the movable range without involving a cost increase.

(6) The present disclosure proposes a multi-joint device including at least two joint mechanisms, in which the at least two joint mechanisms include a first joint mechanism and a second joint mechanism, and the movable range defining member is mounted on at least one of the first joint mechanism or the second joint mechanism. According to this multi-joint device, a multi-joint device including joint mechanisms having different movable ranges can be provided without involving a cost increase.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of an upper rear side of a multi-joint device according to the present embodiment.

(2) FIG. 2 is a perspective view of an upper front side of the multi-joint device according to the present embodiment.

(3) FIG. 3 is a left side view of the multi-joint device according to the present embodiment.

(4) FIG. 4 is a rear view of the multi-joint device according to the present embodiment.

(5) FIG. 5 is a perspective view of an upper rear side of a first joint mechanism according to the present embodiment.

(6) FIG. 6 is a perspective view of an upper front side of the first joint mechanism according to the present embodiment.

(7) FIG. 7 is a rear view of the first joint mechanism according to the present embodiment.

- (8) FIG. 8 is a perspective view of a second bevel gear, a third bevel gear, and the periphery thereof in the first joint mechanism according to the present embodiment.
- (9) FIG. 9 is a left side view of a first bevel gear and the periphery thereof according to the present embodiment.
- (10) FIG. 10 is a cross-sectional view taken along line X-X in FIG. 9.
- (11) FIG. 11 is a perspective view of the first bevel gear according to the present embodiment.
- (12) FIG. 12 is a perspective view of a movable range defining member mounted on a regulated part illustrated in FIG. 11.
- (13) FIG. 13 is a left side view of a regulated part that rotates about an axis ax12, a regulating part that regulates rotation of the regulated part, and the periphery thereof.
- (14) FIG. 14 is a left side view of the movable range defining member mounted on the regulating part illustrated in FIG. 13.

DESCRIPTION OF EMBODIMENT

(15) Hereinafter, an embodiment of the present disclosure will be explained with reference to the drawings.

General Overview of Entire Configuration

- (16) First, the general overview of the entire configuration of a multi-joint device 1 according to the present embodiment will be explained with reference to FIGS. 1 to 4. FIG. 1 is a perspective view of an upper rear side of the multi-joint device according to the present embodiment. FIG. 2 is a perspective view of an upper front side of the multi-joint device according to the present embodiment. FIG. 3 is a left side view of the multi-joint device according to the present embodiment. FIG. 4 is a rear view of the multi-joint device according to the present embodiment.
- (17) In the following explanation, an arrow X1, an arrow X2, an arrow Y1, an arrow Y2, an arrow Z1, and an arrow Z2 in the drawings indicate the right side, the left side, the front side, the rear side, the upper side, and the lower side, respectively. In FIGS. 1 to 4, an axis ax11 and an axis ax22 extend in the front-rear direction, an axis ax12 and an axis ax32 extend in the left-right direction, and an axis ax21 and an axis ax31 extend in the up-down direction. It is to be noted that none of these axes in the drawings exists in reality, and these axes are rotation centers for respective parts, which will be explained later. In the following explanation, the attitude of the multi-joint device 1 illustrated in FIGS. 1 to 4, is referred to as a “basic attitude.”
- (18) In the present embodiment, the multi-joint device 1 includes a first joint mechanism 10, a second joint mechanism 20, and a third joint mechanism 30.
- (19) The first joint mechanism 10 includes a connection object part 11 at one end thereof, and a connecting part 12 at the other end thereof. In the first joint mechanism 10, the connection object part 11 is rotated relatively to the connecting part 12 about the axis ax11 and about the axis ax12 which extends in a direction perpendicular to the axis ax11. Thus, the first joint mechanism 10 is a generally-called biaxial integrated type.
- (20) Likewise, the second joint mechanism 20 includes a connection object part 21 at one end thereof, and a connecting part 22 at the other end thereof. In the second joint mechanism 20, the connection object part 21 is rotated relatively to the connecting part 22 about the axis ax21 and about the axis ax22 which extends in a direction perpendicular to the axis ax21. Thus, the second joint mechanism 20 is a generally-called biaxial integrated type. Likewise, the third joint mechanism 30 includes a connection object part 31 at one end thereof, and a connecting part 32 at the other end thereof. In the third joint mechanism 30, the connection object part 31 is rotated relatively to the connecting part 32 about the axis ax31 and about the axis ax32 which extends in a direction perpendicular to the axis ax31. Thus, the third joint mechanism 30 is a generally-called biaxial integrated type.
- (21) It is to be noted that the term “rotate” in the present description means an action of making a circular movement in either a clockwise direction or a counterclockwise direction about an axis at a circular movement angle of less than 360 degrees.

(22) In FIGS. 1 to 4, the connecting part **12** of the first joint mechanism **10** is connected to the connection object part **21** of the second joint mechanism **20**, and the connecting part **22** of the second joint mechanism **20** is connected to the connection object part **31** of the third joint mechanism **30**. In this structure, rotation of the connection object part **21** relative to the connecting part **22** results in rotation of the first joint mechanism **10** relative to the second joint mechanism **20**. In addition, rotation of the connection object part **31** relative to the connecting part **32** results in rotation of the first joint mechanism **10** and the second joint mechanism **20** relative to the third joint mechanism.

(23) [Joint Mechanism]

(24) Next, with reference to FIGS. 5 to 10, the details of the first joint mechanism **10** will be explained to exemplify a joint mechanism according to the present embodiment. It is to be noted that a detailed explanation of the second joint mechanism **20** and the third joint mechanism **30** will be omitted because these joint mechanisms have the same configuration and operate in the same manner as the first joint mechanism **10**. It is to be noted that FIGS. 5 to 10 each illustrate the first joint mechanism **10** in a state where the multi-joint device **1** is in the basic attitude which is illustrated in FIG. 1, etc.

(25) FIG. 5 is a perspective view of the upper rear side of the first joint mechanism according to the present embodiment. FIG. 6 is a perspective view of the upper front side of the first joint mechanism according to the present embodiment. FIG. 7 is a rear view of the first joint mechanism according to the present embodiment. It is to be noted that a gear cover **C** which is illustrated in FIGS. 1 to 4, is not illustrated in FIGS. 5 to 7 in order to represent bevel gears, which will be explained later. It is to be noted that the gear covers **C** are intended to inhibit the bevel gears from being viewed from the outside, and to protect the bevel gears against dust or the like.

(26) FIG. 8 is a perspective view of a second bevel gear, a third bevel gear, and the periphery thereof in the first joint mechanism according to the present embodiment. FIG. 9 is a left side view of a first bevel gear and the periphery thereof. FIG. 10 is a cross-sectional view taken along line X-X in FIG. 9.

(27) The first joint mechanism **10** mainly includes a base part **110** and a rotary part **120**. The rotary part **120** rotates relatively to the base part **110**. Specifically, in the first joint mechanism **10**, the rotary part **120** rotates when the base part **110** is fixed to a member that is external to the first joint mechanism **10**, and the base part **110** rotates when the rotary part **120** is fixed to a member that is external to the first joint mechanism **10**. The following explanation describes an example in which the rotary part **120** rotates while the base part **110** is fixed to a member that is external to the first joint mechanism **10**.

(28) The first joint mechanism **10** further includes a support part **130**. Incidentally, when the rotary part **120** rotates about the axis ax11, the support part **130** becomes a portion of the base part **110**, and, when the rotary part **120** rotates about the axis ax12, the support part **130** becomes a portion of the rotary part **120**, which will be explained in detail later. Specifically, when the rotary part **120** rotates about the axis ax11, the support part **130** rotates relative to the rotary part **120**, and, when the rotary part **120** rotates about the axis ax12, the support part **130** rotates relative to the base part **110**.

(29) [Joint Mechanism: Rotary Part **120**]

(30) The rotary part **120** includes the connection object part **11** and a first bevel gear **G1**, as illustrated in FIG. 9. The connection object part **11** and the first bevel gear **G1** are integrally formed. Therefore, the connection object part **11** rotates together with the first bevel gear **G1**. It is to be noted that illustration of the connection object part **11** is omitted in FIGS. 5 and 6.

(31) The rotation center of the first bevel gear **G1** is the axis ax11. A bearing hole **h1** through which a first support shaft **131** of the support part **130**, which will be explained later, is inserted is formed in the center portion of the first bevel gear **G1**. The bearing hole **h1** may be formed to reach the rear surface side of the first bevel gear **G1** (see FIG. 6).

(32) [Joint Mechanism: Base Part **110**]

(33) The base part **110** includes an exterior housing **H** that forms the exterior of the first joint mechanism **10**, a second bevel gear **G2**, and a third bevel gear **G3**. The rotation center of the second bevel gear **G2** and the third bevel gear **G3** is the axis **ax12**.

(34) The second bevel gear **G2** is disposed so as to be engaged with the first bevel gear **G1**, and to rotate about the axis **ax12**, as illustrated in FIGS. **5**, **10**, or the like.

(35) The second bevel gear **G2** is formed integrally with a spur gear **SG1**, as illustrated in FIGS. **8** and **10**. Upon receiving a driving force from a first motor **M1** which is a driving source, the spur gear **SG1** rotates. The second bevel gear **G2** rotates together with the spur gear **SG1**. A driving force from the first motor **M1** may be inputted to the spur gear **SG1** via a speed reduction gear or the like. The first motor **M1** may be housed in the exterior housing **H**. It is to be noted that a detailed explanation of the structure and arrangement of the first motor **M1** will be omitted. The same applies to a second motor **M2**, which will be explained later.

(36) A bearing hole **h2** through which a second support shaft **132** of the support part **130**, which will be explained later, is inserted is formed in the center portion of the second bevel gear **G2**. The bearing hole **h2** is formed so as to reach the rear surface side of the spur gear **SG1** (see FIG. **8**).

(37) The third bevel gear **G3** is engaged with the first bevel gear **G1**, as illustrated in FIGS. **7**, **10**, or the like. Further, the third bevel gear **G3** is disposed so as to face the second bevel gear **G2** in the axis **ax12** direction, and rotates about the axis **ax12**.

(38) The third bevel gear **G3** is formed integrally with a spur gear **SG2**, as illustrated in FIGS. **8** and **10**. Upon receiving a driving force from a second motor **M2** which is a driving source that is different from the first motor **M1**, the spur gear **SG2** rotates. The third bevel gear **G3** rotates together with the spur gear **SG2**. It is to be noted that a driving force from the second motor **M2** may be inputted to the spur gear **SG2** via a speed reduction gear or the like.

(39) A bearing hole **h3** through which a second support shaft **132** of the support part **130**, which will be explained later, is inserted is formed in the center portion of the third bevel gear **G3** (see FIG. **8**).

(40) It is to be noted that FIG. **10** or the like illustrates the second bevel gear **G2** including two radially split members. However, this is just an example. The second bevel gear **G2** may be formed of one member. The same applies to the third bevel gear **G3**.

(41) [Joint Mechanism: Support Part **130**]

(42) The support part **130** includes the first support shaft **131** extending along the axis **ax11**, and the second support shaft **132** extending along the axis **ax12**. The first support shaft **131** and the second support shaft **132** each have a columnar shape or a cylindrical shape.

(43) The support part **130** supports the first bevel gear **G1**, the second bevel gear **G2**, and the third bevel gear **G3**. More specifically, the first support shaft **131** of the support part **130** is inserted through the bearing hole **h1** formed in the first bevel gear **G1**, and supports the first bevel gear **G1**. Also, the second support shaft **132** of the support part **130** is inserted through the bearing hole **h2** formed in the second bevel gear **G2** and the bearing hole **h3** formed in the third bevel gear **G3**, and supports the second bevel gear **G2** and the third bevel gear **G3**. It is to be noted that a bearing or the like (not illustrated) is preferably disposed between the outer circumferential surface of the first support shaft **131** and the inner circumferential surface of the bearing hole **h1** in the first bevel gear **G1**. Likewise, bearings or the like are preferably disposed between the outer circumferential surface of the second support shaft **132** and the inner circumferential surface of the bearing hole **h2** in the second bevel gear **G2** and between the outer circumferential surface of the second support shaft **132** and the inner circumferential surface of the bearing hole **h3** in the third bevel gear **G3**.

(44) The support part **130** is separate from the first bevel gear **G1**, the second bevel gear **G2**, and the third bevel gear **G3**. Specifically, the first bevel gear **G1** is supported by the first support shaft **131** in such a way that the inner circumferential surface of the bearing hole **h1** in the first bevel gear **G1** is rotatable about the axis **ax11** relatively to the outer circumferential surface of the first

support shaft **131**. Also, the second bevel gear **G2** is supported by the second support shaft **132** in such a way that the inner circumferential surface of the bearing hole **h2** in the second bevel gear **G2** is rotatable about the axis **ax12** relatively to the outer circumferential surface of the second support shaft **132**. In addition, the third bevel gear **G3** is supported by the second support shaft **132** in such a way that the inner circumferential surface of the bearing hole **h3** in the third bevel gear **G3** is rotatable about the axis **ax12** relatively to the outer circumferential surface of the second support shaft **132**.

(45) The second support shaft **132** is supported by a bearing (not illustrated) that is disposed on an upper right part **HR** of the exterior housing **H** and an upper left part **HL** of the exterior housing **H** so as to connect the upper right part **HR** and the upper left part **HL**, as illustrated in FIG. 5, or the like. It is to be noted the upper right part **HR** includes a cover part **C2** that covers the spur gear **SG2**, as illustrated in FIGS. 5 and 6. The cover part **C2** is a part of the exterior housing **H**, and has a bottomed tubular shape formed along the outer shape of the spur gear **SG2**. The upper left part **HL** also includes a cover part **C1** that covers the spur gear **SG1**. The cover part **C1** is a part of the exterior housing **H**, and has a bottomed tubular shape formed along the outer shape of the spur gear **SG1**.

(46) The second support shaft **132** passes through the bearing hole **h2**, and is projected from the rear surface side of the spur gear **SG1**, as illustrated in FIG. 8 or the like. In addition, a regulated part **B2** extending in a direction perpendicular to the axis **ax12** is mounted on a projection that is a portion, of the second support shaft **132**, projected from the rear surface side of the spur gear **SG1**. The regulated part **B2** is mounted on the second support shaft **132** so as to rotate together with rotation of the second support shaft **132** about the axis **ax12**. It is to be noted that the details of the regulated part **B2** will be explained later.

(47) In addition, the support part **130** includes an intervening section **135** that is a section where the first support shaft **131** and the second support shaft **132** intersect with each other, and intervenes between the second bevel gear **G2** and the third bevel gear **G3**, as illustrated in FIGS. 5, 8, or the like. With the intervening section **135**, the second bevel gear **G2** and the third bevel gear **G3** are positioned with respect to the extension direction of the axis **ax12**. The intervening section **135** is formed integrally with the first support shaft **131** and the second support shaft **132**.

(48) Further, the intervening section **135** includes a diameter enlarged section **135a** that has a larger diameter than the first support shaft **131** and is integrated with the first support shaft **131**, as illustrated in FIG. 8. A regulating part **A1** extending in a direction perpendicular to the axis **ax11** is formed on the outer circumferential surface of the diameter enlarged section **135a**. It is to be noted that the details of the regulating part **A1** will be explained later.

(49) It is to be noted that, on a side opposite to the diameter enlarged section **135a** in the extension direction of the axis **ax11**, the gear cover **C** which is illustrated in FIG. 1, etc., but is not illustrated in FIGS. 5 to 10, is preferably fixed to the intervening section **135**.

(50) [Joint Mechanism: Rotation about Axis **Ax11**]

(51) As previously explained, the second bevel gear **G2** rotates upon receiving a driving force outputted from the first motor **M1**, while the third bevel gear **G3** rotates upon receiving a driving force outputted from the second motor **M2**.

(52) If the second bevel gear **G2** rotates clockwise with respect to the **X2** direction, a force **F11** acts upwardly in a portion, of the first bevel gear **G1**, engaged with the second bevel gear **G2** (see FIG. 10). If the third bevel gear **G3** rotates counterclockwise with respect to the **X2** direction, a force **F22** acts downwardly in a portion, of the first bevel gear **G1**, engaged with the third bevel gear **G3** (see FIG. 10). With the force **F11** and the force **F22**, the first bevel gear **G1** rotates about the axis **ax11** to an **R1** direction (clockwise direction) which is illustrated in FIG. 10. According to the **R1** direction rotation of the first bevel gear **G1**, the rotary part **120** rotates clockwise with respect to the **Y2** direction.

(53) If the second bevel gear **G2** rotates counterclockwise with respect to the **X2** direction, a force

F12 acts downwardly in a portion, of the first bevel gear **G1**, engaged with the second bevel gear **G2** (see FIG. 10). If the third bevel gear **G3** rotates clockwise with respect to the **X2** direction, a force **F21** acts upwardly in a portion, of the first bevel gear **G1**, engaged with the third bevel gear **G3** (see FIG. 10). With the force **F12** and the force **F21**, the first bevel gear **G1** rotates about the axis **ax11** to an **R2** direction (counterclockwise direction) which is illustrated in FIG. 10. According to the **R2** direction rotation of the first bevel gear **G1**, the rotary part **120** rotates counterclockwise with respect to the **Y2** direction.

(54) Thus, if the second bevel gear **G2** and the third bevel gear **G3** rotate oppositely with respect to the same direction, the first bevel gear **G1** rotates about the axis **ax11**. As a result, the rotary part **120** rotates about the axis **ax11** relatively to the base part **110**.

(55) [Joint Mechanism: Rotation about Axis **Ax12**]

(56) If the second bevel gear **G2** rotates clockwise with respect to the **X2** direction, the force **F11** acts upwardly in a portion, of the first bevel gear **G1**, engaged with the second bevel gear **G2** (see FIG. 10). If the third bevel gear **G3** rotates clockwise with respect to the **X2** direction, the force **F21** acts upwardly in a portion, of the first bevel gear **G1**, engaged with the third bevel gear **G3** (see FIG. 10). With these upward forces **F11** and **F21**, the first bevel gear **G1** rotates about the axis **ax12** so as to follow the shape of the second bevel gear **G2** and the shape of the third bevel gear **G3**. Accordingly, the rotary part **120** rotates about the axis **ax12** to an **R3** direction which is illustrated in FIG. 9.

(57) If the second bevel gear **G2** rotates counterclockwise with respect to the **X2** direction, the force **F12** acts downwardly in a portion, of the first bevel gear **G1**, engaged with the second bevel gear **G2** (see FIG. 10). If the third bevel gear **G3** rotates counterclockwise with respect to the **X2** direction, the force **F22** acts downwardly in a portion, of the first bevel gear **G1**, engaged with the third bevel gear **G3** (see FIG. 10). With these downward forces **F12** and **F22**, the first bevel gear **G1** rotates about the axis **ax12** so as to follow the shape of the second bevel gear **G2** and the shape of the third bevel gear **G3**. Accordingly, the rotary part **120** rotates about the axis **ax12** to an **R4** direction which is illustrated in FIG. 9.

(58) In addition, when the rotary part **120** including the first bevel gear **G1** rotates about the axis **ax12** to the **R3** or **R4** direction illustrated in FIG. 9, the outer circumferential surface of the first support shaft **131** of the support part **130** is pushed against the inner circumferential surface of the bearing hole **h1** in the first bevel gear **G1**. Accordingly, the support part **130** rotates about the axis **ax12**.

(59) [Movable Range of Joint Mechanism]

(60) The present embodiment adopts a configuration of physically regulating the rotational movable ranges of the first joint mechanism **10** about the axis **ax11** and the axis **ax12**. More specifically, the rotational movable range about the axis **ax11** is regulated by the regulating part **A1**, and the rotational movable range about the axis **ax12** is regulated by a regulating part **A2**. A more specific explanation will be given below.

(61) [Movable Range of Joint Mechanism: Rotation about Axis **Ax11**]

(62) FIG. 11 is a perspective view of a first bevel gear according to the present embodiment.

(63) In the first bevel gear **G1**, a center hole **h11** having a larger diameter than the bearing hole **h1** through which the first support shaft **131** is inserted, is formed coaxially with the bearing hole **h1**. A regulated part **B1** is disposed on the inner circumferential surface of the center hole **h11**. The regulated part **B1** rotates about the axis **ax11** together with rotation of the first bevel gear **G1** about the axis **ax11**.

(64) The regulating part **A1** is disposed on the first support shaft **131** of the support part **130**, as illustrated in FIG. 8 or the like. The regulating part **A1** is disposed on the extension of the rotation locus of the regulated part **B1**. The regulating part **A1** has a function of regulating rotation of the regulated part **B1** within a first movable range **Ra1** which is illustrated in FIG. 10.

(65) For example, if the first bevel gear **G1** rotates about the axis **ax11** to the **R1** direction which is

illustrated in FIG. 10, the regulated part B1 rotates to the R1 direction together with the rotation of the first bevel gear G1. After rotating by a prescribed angle about the axis ax11, the regulated part B1 comes into contact with the regulating part A1. When the regulated part B1 is caught on the regulating part A1 in this manner, the rotation of the regulated part B1 to the R1 direction is regulated.

(66) If the first bevel gear G1 rotates about the axis ax11 to the R2 direction which is illustrated in FIG. 10, the regulated part B1 rotates to the R2 direction together with the rotation of the first bevel gear G1. After rotating by a prescribed angle about the axis ax11, the regulated part B1 comes into contact with the regulating part A1. When the regulated part B1 is caught on the regulating part A1 in this manner, the rotation of the regulated part B1 to the R2 direction is regulated.

(67) Rotation of the regulated part B1 about the axis ax11 is regulated in this manner, so that rotation of the rotary part 120 about the axis ax11 relative to the base part 110 is regulated.

(68) In the first joint mechanism 10 according to the present embodiment, a movable range defining member RA1 is mounted on the regulated part B1 of the first bevel gear G1. FIG. 12 is a perspective view of a movable range defining member mounted on the regulated part illustrated in FIG. 11.

(69) The movable range defining member RA1 is configured to set the movable range of the regulated part B1 to be narrower than the first movable range Ra1 which is illustrated in FIG. 10. The movable range defining member RA1 is separate from the regulated part B1, and is preferably attachable to/detachable from the regulated part B1.

(70) FIG. 12 illustrates an example in which the movable range defining member RA1 is formed so as to almost completely cover the regulated part B1, and has a length in the circumferential direction of the axis ax11, longer than the regulated part B1. However, the movable range defining member RA1 is not limited to this example. It is sufficient to dispose the movable range defining member RA1 in such a way that the movable range of the regulated part B1 becomes narrower than at least the first movable range Ra1 illustrated in FIG. 10. For example, the movable range defining member RA1 may be mounted to cover only one end side, in the circumferential direction of the axis ax11, of the regulated part B1. The movable range defining member RA1 is not necessarily formed to cover the regulated part B1, and thus, the movable range defining member RA1 may be disposed apart from the regulated part B1 in the circumferential direction of the axis ax11. In this case, it is preferable that the movable range defining member RA1 is fixedly mounted on the inner circumferential surface of the center hole h11 in the first bevel gear G1.

(71) In the present embodiment, since the movable range defining member RA1 is used, the movable range of rotation of the rotary part 120 about the axis ax11 can be controlled for the purpose of use of the first joint mechanism 10, etc. Therefore, as a result of use of the first joint mechanism 10 according to the present embodiment, it is not necessary to prepare joint mechanisms having different movable ranges according to respective purposes of use. Consequently, a cost increase can be suppressed.

(72) Further, since the regulated part B1 is configured to come into contact with the regulating part A1 according to rotation about the axis ax11, the regulated part B1 may be worn out and deteriorate after a long-term use. In the present embodiment, the movable range defining member RA1 is mounted so as to cover the regulated part B1, so that the regulated part B1 is inhibited from being worn out or deteriorating. That is, the movable range defining member RA1 additionally has a function of protecting the regulated part B1. If the movable range defining member RA1 is worn out and deteriorates, the movable range defining member RA1 can be removed from the regulated part B1, and be replaced with a new one. Since the movable range defining member RA1 is replaceable, it is not necessary to replace the first joint mechanism 10 itself even when the movable range defining member RA1 is worn out and deteriorates. Thus, a cost increase can be suppressed. It is to be noted that it is sufficient that the movable range defining member RA1 is separate from at least the regulated part B1. The movable range defining member RA1 include the same material

as that of the regulated part B1, or may include a material different from the material of the regulated part B1.

(73) [Movable Range of Joint Mechanism: Rotation about Axis ax12]

(74) FIG. 13 is a left side view of a regulated part that rotates about the axis ax12, a regulating part that regulates rotation of the regulated part, and the periphery thereof.

(75) The regulating part A2 is disposed in the upper left part HL (cover part C2) of the exterior housing H of the first joint mechanism 10. The regulating part A2 is disposed on the extension of the rotation locus of the supported part B2. The regulating part A2 has a function of regulating rotation of the regulated part B2 within a prescribed movable range. It is preferable that the regulating part A2 has an arc shape following the shape of the upper left part HL (cover part C2), as illustrated in FIG. 5 or the like. The regulating part A2 regulates the movable range of the regulated part B2 according to the length of the regulating part A2 in the circumferential direction of the axis ax12. Specifically, when the regulating part A2 is long in the circumferential direction of the axis ax12, the movable range of the regulated part B2 is narrow. When the regulating part A3 is short in the circumferential direction of the axis ax12, the movable range of the regulated part B2 is wide.

(76) For example, if the first bevel gear G1 rotates about the axis ax12 to the R3 direction which is illustrated in FIG. 9, the regulated part B2 rotates to the R3 direction together with the rotation of the first bevel gear G1. Then, after rotating by a prescribed angle about the axis ax12, the regulated part B2 comes into contact with the regulating part A2. When the regulated part B2 is caught on the regulating part A2 in this manner, the rotation of the regulated part B2 to the R3 direction is regulated.

(77) If the first bevel gear G1 rotates about the axis ax12 to the R4 direction which is illustrated in FIG. 9, the regulated part B2 rotates to the R4 direction together with the rotation of the first bevel gear G1. After rotating by a prescribed angle about the axis ax12, the regulated part B2 comes into contact with the regulating part A2. When the regulated part B2 is caught on the regulating part A2 in this manner, the rotation of the regulated part B2 to the R4 direction is regulated.

(78) As explained so far, rotation of the regulated part B2 about the axis ax12 is regulated, whereby rotation of the rotary part 120 relative to the base part 110 about the axis ax12 is regulated.

(79) In the first joint mechanism 10 according to the present embodiment, a movable range defining member RA2 is mounted on the regulating part A2. FIG. 14 is a left side view of a movable range defining member mounted on the regulating part illustrated in FIG. 13.

(80) The movable range defining member RA2 sets the movable range of the regulated part B2 to a second movable range Ra4 that is narrower than a first movable range Ra3 which is illustrated in FIG. 13. The movable range defining member RA2 is separate from the regulating part A2, and is preferably attachable to/detachable from the regulating part A2.

(81) FIG. 14 illustrates an example in which the movable range defining member RA2 is mounted on one end side of the regulating part A2 in the circumferential direction of the axis ax12.

However, the movable range defining member RA2 is not limited to this example. It is sufficient to dispose the movable range defining member RA2 in such a way that the movable range of the regulated part B2 becomes narrower than at least the first movable range Ra3 illustrated in FIG. 13. For example, the movable range defining members RA2 may be mounted on both ends of the regulating part A2 in the circumferential direction of the axis ax12. Alternatively, the movable range defining member RA2 may be disposed apart from the regulating part A2 in the circumferential direction of the axis ax12. In this case, on the extension of the rotation locus of the regulated part B2 about the axis ax12, the movable range defining member RA2 may be fixed to the exterior housing H of the base part 110, for example.

(82) In the present embodiment, since the movable range defining member RA2 is used, the movable range of rotation of the rotary part 120 about the axis ax12 can be controlled according to the purpose of use of the first joint mechanism 10, etc. Therefore, as a result of use of the first joint mechanism 10 according to the present embodiment, it is not necessary to prepare joint

mechanisms having different movable ranges according to respective purposes of use.

Consequently, a cost increase can be suppressed.

(83) Further, since the regulated part **B2** comes into contact with the regulating part **A2** according to rotation of the regulated part **B2** about the axis **ax12**, the regulating part **A2** may be worn out and deteriorate after a long-term use. In the present embodiment, since the movable range defining member **RA2** is mounted so as to cover a portion of the regulating part **A2** with which the regulated part **B2** comes into contact, the regulating part **A2** is inhibited from being worn out or deteriorating. That is, the movable range defining member **RA2** additionally has a function of protecting the regulating part **A2**. If the movable range defining member **RA2** is worn out and deteriorates, the movable range defining member **RA2** may be removed from the regulating part **A2**, and be replaced with a new one. Since the movable range defining member **RA2** is replaceable, it is not necessary to replace the first joint mechanism **10** itself even when the movable range defining member **RA2** is worn out and deteriorates. Thus, a cost increase can be suppressed.

(84) [Others]

(85) The first joint mechanism **10** has been explained with reference to FIGS. **5** to **14**. It is preferable that the second joint mechanism **20** and the third joint mechanism **30** also have the same configuration. That is, it is preferable that the first joint mechanism **10**, the second joint mechanism **20**, and the third joint mechanism **30** are identical to each other. Further, the presence/absence of the movable range defining members **RA1** and **RA2** is preferably decided according to the purpose of use or according to a position. In the multi-joint device **1** illustrated in FIGS. **1** to **4**, the movable range defining member **RA2** is mounted on the regulating part **A2** in each of the first joint mechanism **10** and the third joint mechanism **30**, while no movable range defining member is mounted on the regulating part **A2** of the second joint mechanism **20**. In this configuration, the rotational movable range of the first joint mechanism **10** about the axis **ax12** is equal to the rotational movable range of the third joint mechanism **30** about the axis **ax32**. On the other hand, the rotational movable range of the second joint mechanism **20** about the axis **ax22** is wider than the rotational movable range of the first joint mechanism **10** about the axis **ax12**. It is to be noted that the presence/absence of the movable range defining member **RA1** which is mounted on the regulated part **B1** of the first bevel gear **G1** is also preferably decided according to the purpose of use or according to a position. This is not illustrated in the drawings though.

(86) It is to be noted that the multi-joint device **1** is preferably used for a human or animal-form robot that can walk on two legs or four legs, for example. In a case where the multi-joint device **1** including a plurality of joint mechanisms is used for an arm of a human form robot, for example, one of the joint mechanisms is used as a shoulder joint, and another joint mechanism is used as an elbow joint.

(87) The multi-joint device **1** including the three joint mechanisms has been explained in the present embodiment. However, the number of joint mechanisms is not limited, and thus, the number may be one, two, four, or greater.

(88) In the multi-joint device **1** illustrated in FIG. **1**, etc., any other joint mechanism may be additionally connected to the connection object part **11** of the first joint mechanism **10** or the connecting part **32** of the third joint mechanism **30**. Further, an end effector of any type may be additionally connected to the connection object part **11** or the connecting part **32**.

(89) In the above-mentioned embodiment, the second bevel gear **G2** and the third bevel gear **G3** rotate in the same direction or in the opposite directions, when viewed from the same direction. Alternatively, either the second bevel gear **G2** or the third bevel gear **G3** may be driven to rotate while the other gear is stopped. In this case, the rotary part **120** rotates relatively to the base part **110** so as to be twisted with respect to the axis **ax11** and the axis **ax12**, but a detailed explanation thereof will be omitted.

(90) In addition, it is preferable that the regulating part **A2** is formed so as to come into surface-contact with the regulated part **B2**. If so, in a state where the movable range defining member **RA2**

is not mounted, the regulating part **A2** receives, by a surface thereof, a shock caused by the contact with the regulated part **B2**, and a local force is inhibited from acting on the regulating part **A2**. As a result, the durability of the regulating part **A2** is improved, and the life of the regulating part **A2** is increased. In addition, it is preferable that the movable range defining member **RA2** is also formed so as to come into surface-contact with the regulated part **B2**. If so, the movable range defining member **RA2** receives, by a surface thereof, a shock caused by the contact with the regulated part **B2**, and a local force is inhibited from acting on the movable range defining member **RA2**. As a result, the durability of the movable range defining member **RA2** is improved, and the life of the movable range defining member **RA2** is increased.

(91) In the present embodiment, the movable range defining member **RA2** is mounted on the regulating part **A2** that is disposed on the rear surface side of the second bevel gear **G2** and the spur gear **SG1**, whereby an access to the movable range defining member **RA2** from the outside can be easily made. That is, it is easy to attach/detach the movable range defining member **RA2** to/from the regulating part **A2**.

(92) In the above-mentioned embodiment, the regulated part **B1** is disposed on the inner circumferential surface of the center hole **h11** in the first bevel gear **G1**. Alternatively, the regulated part **B1** may be disposed on the rear surface side of the first bevel gear **G1**. If so, the regulating part **A1** is preferably disposed on a projection that is a portion, of the first support shaft **131**, projected from the rear surface side of the first bevel gear **G1**. In this case, however, it is difficult to connect another member to the connection object part **11** of the rotary part **120**. That is, when the configuration of the present embodiment in which the regulated part **B1** is disposed on the inner circumferential surface of the center hole **h11** in the first bevel gear **G1** is adopted, a structure in which another member can be easily connected to the rotary part **120** of the first joint mechanism **10**, is realized.

(93) In the present embodiment, the movable range defining member **RA1** is mounted on the regulated part **B1**. Alternatively, the movable range defining member **RA1** may be mounted on the regulating part **A1**. Also, in the present embodiment, the movable range defining member **RA2** is mounted on the regulating part **A2**. Alternatively, the movable range defining member **RA2** may be mounted on the regulated part **B2**.

(94) Incidentally, in the present embodiment, the base part **110** is fixed while the rotary part **120** rotates. However, it is sufficient that the rotary part **120** rotates relatively to the base part **110**. That is, if the rotary part **120** is fixed, the base part **110** may rotate. In this case, the regulated part **B1** functions as a member for regulating rotation of the regulating part **A1**, while the regulated part **B2** functions as a member for regulating rotation of the regulating part **A2**.

Claims

1. A joint mechanism including a base part and a rotary part that rotates relatively to the base part, the joint mechanism comprising: a regulated part that is disposed on an inner circumferential surface of the rotary part and rotates together with the rotary part, wherein the rotary part includes a first bevel gear that rotates about a first axis direction; a regulating part that is disposed on an extension of a rotation locus of the regulated part, and has a function of regulating rotation of the regulated part relative to the base part, within a first movable range, wherein the base part includes a second bevel gear that is engaged with the first bevel gear and that rotates about a second axis direction intersecting with the first axis direction, and a third bevel gear that is engaged with the first bevel gear, is disposed so as to face the second bevel gear with respect to an extension direction of the second axis direction, and rotates about the second axis direction; a second regulated part that rotates about the second axis direction relatively to the base part; a second regulating part that is disposed on an extension of a rotation locus of the second regulated part; and a movable range defining member that is mounted on either the base part or the rotary part, and

defines, as a movable range of the regulated part, a second movable range that is narrower than the first movable range.

2. The joint mechanism according to claim 1, wherein the movable range defining member is mounted on either the regulated part or the regulating part.

3. The joint mechanism according to claim 1, wherein the inner circumferential surface is of a center hole that is formed coaxially with the first axis direction in the first bevel gear, and the regulating part is disposed on a first support shaft that supports the first bevel gear.

4. The joint mechanism according to claim 3, wherein a connection object part to which a member that is external to the joint mechanism is connected is disposed on a rear surface side of the first bevel gear in the rotary part.

5. The joint mechanism according to claim 4, further comprising: a second support shaft that supports the second bevel gear and the third bevel gear, wherein the second regulated part is disposed on the second support shaft, and the second regulating part is disposed on the base part.

6. The joint mechanism according to claim 5, wherein the second regulated part is mounted on a projection that is a portion, of the second support shaft, projected from the rear surface side of the second bevel gear.

7. The joint mechanism according to claim 3, wherein the movable range defining member is mounted on at least the regulated part.

8. The joint mechanism according to claim 3, wherein the joint mechanism further comprises a second movable range defining member that is mounted on at least the second regulating part.

9. A multi-joint device comprising: at least two joint mechanisms each including a base part and a rotary part that rotates relatively to the base part, the joint mechanism including a regulated part that is disposed on an inner circumferential surface of the rotary part and rotates together with the rotary part, wherein the rotary part includes a first bevel gear that rotates about a first axis direction; a regulating part that is disposed on an extension of a rotation locus of the regulated part, and has a function of regulating rotation of the regulated part relative to the base part, within a first movable range, wherein the base part includes a second bevel gear that is engaged with the first bevel gear and that rotates about a second axis direction intersecting with the first axis direction, and a third bevel gear that is engaged with the first bevel gear, is disposed so as to face the second bevel gear with respect to an extension direction of the second axis direction, and rotates about the second axis direction, a second regulated part that rotates about the second axis direction relatively to the base part, a second regulating part that is disposed on an extension of a rotation locus of the second regulated part, and a movable range defining member that is mounted on either the base part or the rotary part, and defines, as a movable range of the regulated part, a second movable range that is narrower than the first movable range, wherein: the at least two joint mechanisms include a first joint mechanism and a second joint mechanism, and the movable range defining member is mounted on at least one of the first joint mechanism or the second joint mechanism.
